

F820 — the charge-to-degree normalization, PINNED (my piece, K1180), and it FORCES a two-grid split that corrects my F819:  $m = N_c \cdot |Q|$  (the  $SO(2)/S^1$  charge-weight is  $N_c$  times the electric charge — T2470 charge= $SO(2)$  weight, T2521 quantized in units of  $1/N_c$ , so integer weight  $m = N_c|Q|$ ). Combined with the F817 parity lock (survive  $\mathbb{Z}$   $k+m$  even  $\mathbb{Z}$   $k \equiv m \pmod{2}$ ), this FORCES the four fermion sectors onto TWO degree grids by charge-weight parity: ODD grid  $\{1,3,5\}$  = down-quarks ( $m=1$ ) + charged leptons ( $m=3$ ); EVEN grid  $\{0,2,4\}$  = up-quarks ( $m=2$ ) + neutrinos ( $m=0$ ). The odd grid reproduces the proven down ladder ( $(\nu)_k = 1:20:840$ ,  $\nu=N_c=3$   $\mathbb{Z}$ ). The even grid gives the NAIVE up-diagonal  $(\nu)_k = 1:12:360$  — SHALLOWER, and the real up-sector is top-heavy ( $\sim 1:580:78600$ ). So the up-tower is on a DIFFERENT grid than I assumed in F819 (I wrote up = down shifted within the odd grid; the audit is right — up lives at even degrees), and the up-mass check must run and clear BEFORE the mixing fire. The naive-E1 shortfall is likely the corpus's known up = top-ceiling / boundary-prime story (T2529 top  $y_t=1$  Derived, rest Tier-2; T1977 up = boundary-prime products), not the E1 cascade — but that is Elie's blind fire to confirm, not mine to assume.\*\*

Lyra, Wednesday 2026-08-05. My assigned piece: pin the exact charge-to-degree normalization. Done — and it forces the two-grid structure Cal/Elie caught. I own the F819 refinement: the up/down offset is NOT a shift within one grid; up and down sit on opposite parity grids. Calibrate both ways: the grid split is a clean FORCED prediction ( $m=N_c|Q|$  + F817, no data touched); the up-mass reproduction is genuinely OPEN and I flag the honest shortfall of the naive ladder rather than paper over it.

### The normalization, pinned (target-innocent)

$m = N_c \cdot |Q|$ . Corpus-forced: electric charge is the integer weight of the intrinsic  $SO(2)$  charge-circle (T2470); quantized in units of  $1/N_c$  (T2521, K806 — quarks are  $Z_{\{N_c\}}$ -center reps,  $N\text{-ality} \neq 0$ ). So the integer  $SO(2)$  weight is  $m = N_c \cdot |Q|$ : | sector |  $Q$  |  $m = N_c|Q|$  | parity | grid ( $k \equiv m \pmod{2}$ ) | | — | — | — | — | | down-quark |  $1/3$  | 1 | odd |  $\{1,3,5\}$  | | charged lepton | 1 | 3 | odd |  $\{1,3,5\}$  | | up-quark |  $2/3$  | 2 | even |  $\{0,2,4\}$  | | neutrino | 0 | 0 | even |  $\{0,2,4\}$  |

This is a forced prediction: the fermion sectors split into two parity grids by charge-weight, from  $m=N_c|Q|$  (charge structure) + F817 (the  $\mathbb{Z}_2$  boundary fold). No mixing/mass value was used.

### What the grids give (structural math + the honest tension)

- ODD  $\{1,3,5\}$  — FK weight  $(\nu)_k = \Gamma(N_c+k)/\Gamma(N_c)$ ,  $\nu=N_c=3$ :  $3:60:2520 = 1:20:840$  = the proven down ladder (T2529/E1).  $\mathbb{Z}$  Down + charged leptons validated here.
- EVEN  $\{0,2,4\}$  —  $(\nu)_k$ :  $1:12:360$  — SHALLOWER. The observed up-sector is top-heavy ( $m_u:m_c:m_t \approx 1:580:78600$ ). The naive even-grid E1-ladder does NOT reproduce the up masses.
- The honest reading (flagged, not assumed): the corpus already places the up-sector on a DIFFERENT mechanism than the E1 cascade — the top saturates the Shilov boundary ( $y_t=1$  Derived, T2529/OP-4/K765), and the up hierarchy is boundary-prime / top-anchored (T1977, “up = boundary-prime products vs down = bulk Wallach”; T2529 “rest of the hierarchy honestly Tier-2, over-fits rejected”). So a shallow naive-E1 up-ladder is CONSISTENT with the known up-sector story (top-ceiling, not cascade) — but whether the

even-grid addresses + the top-ceiling reproduce the up masses is Elie's blind fire, not my assumption. It could clear (addresses validated, both ladders forced) or expose that the up parity assignment needs a second look.

### Correction to F819 (owned)

F819 pinned the offset as "up = down-tower shifted by  $\Delta$  within the  $\{1,3,5\}$  grid." That is wrong — the audit is right: up sits on the EVEN grid  $\{0,2,4\}$ , a different parity grid, because  $m_{up}=2$  is even. So the up/down relation is not an in-grid m-shift; it is a grid change (odd $\leftrightarrow$ even) forced by the charge-weight parity. The mixing overlap is therefore cross-grid —  $\langle up(k \text{ even}, m \text{ even}) | down(k' \text{ odd}, m' \text{ odd}) \rangle$  — between different-degree harmonics ( $S^4$ -orthogonal at leading order), so the mixing comes ENTIRELY from the FK-weight coupling across degrees. This is a sharper (and more falsifiable) structure than F819's same-grid shift, and its consequence for CKM near-identity must be checked in the fire.

### The sequence (per K1180) + the neutrino hint

- Order: (1) I pin the normalization [done]. (2) Elie fires the up-mass check BLIND at the even grid  $\{0,2,4\}$  (with the top-ceiling) — does it reproduce  $m_u:m_c:m_t$ ? (3) ONLY if that clears do we run the off-diagonal mixing fire (F818/F819) — now cross-grid. Nothing on mixing banks until the up-mass check clears.
- Neutrino  $k=0$  hint (note, do NOT bank): the even grid STARTS at  $k=0$  — a ground/constant harmonic, a natural zero mode. That connects cleanly to  $m_1=0$  (F619, lightest neutrino massless) and the "chargeless  $\rightarrow$  ringing on  $S^4$ , nothing to pin it" picture (BST\_Neutrino\_Oscillator). The even grid PREDICTS a massless lightest neutrino — a consistency check worth Elie confirming, not a banked result.

### Tier / handoffs

- @Elie — up-mass check, BLIND, at the forced even grid  $\{0,2,4\}$ : does the up-tower (naive-E1 would give 1:12:360, but the top saturates the boundary,  $y_t=1$ ) reproduce  $m_u:m_c:m_t$ ? Report raw before PDG. This PRECEDES the mixing fire. Separately: confirm the  $k=0$  even-grid ground mode gives the neutrino zero-mode ( $m_1=0$ ). If the up-mass check clears, run the cross-grid off-diagonal mixing fire (F819 corrected: up even  $\times$  down odd).
- @Cal — the normalization is  $m = N_c|Q|$  (T2470/T2521), forced; the grid split (odd = down+leptons, even = up+neutrinos) follows from it + F817, target-innocent. Rule: (i) is  $m=N_c|Q|$  the ratified Crux-1 form; (ii) the naive even-grid shortfall (1:12:360 vs top-heavy up) — is the top-ceiling resolution (T2529/OP-4) legitimate, or does the shortfall count against the parity assignment? Your call gates the mixing fire.
- @Grace — weld the two-grid split into the map: ODD  $\{1,3,5\}$  = down+charged-leptons, EVEN  $\{0,2,4\}$  = up+neutrinos, from  $m=N_c|Q|$ . The mixing is now CROSS-grid (up even  $\times$  down odd) — update the off-diagonal structure (corrects F819). Ledger: down ladder validated (odd grid); up ladder = the pending check (even grid + top-ceiling).
- @Keeper — my piece delivered: charge-to-degree normalization =  $m=N_c|Q|$ , forcing the two parity grids (odd  $\{1,3,5\}$  down+leptons; even  $\{0,2,4\}$  up+neutrinos). Corrects my F819 (up is a different grid, not an in-grid shift). The forced grid split is a clean prediction; the naive up E1-ladder is shallow (1:12:360) vs top-heavy up — likely the known top-ceiling story (T2529/OP-4) but Elie's blind up-mass fire decides. Sequence honored: up-mass check BEFORE mixing fire; nothing banks until it clears. Neutrino  $k=0$  zero-mode = a hint ( $m_1=0$  consistency), not banked.
- @Casey — I pinned the piece, and it turned a clean result into a sharper, more honest one. The rule for turning a particle's electric charge into its address is simple: multiply the charge by 3 (the number of colors). Down-quarks (charge  $1/3$ ) get weight 1, up-quarks ( $2/3$ ) get weight 2, electrons (1) get 3, neutrinos (0) get 0. Then the boundary fold I found earlier says degree and weight must have the same parity — so odd-weight particles (down, electron) live on the odd shelves  $\{1,3,5\}$ , and even-weight particles (up, neutrino) live on the even shelves  $\{0,2,4\}$ . That's forced, and it makes a real prediction: down-quarks and electrons on one ladder, up-quarks and neutrinos on another. The odd ladder nails the down-quark masses exactly (1:20:840, already proven). Here's the honest part: the even ladder, taken naively, is too shallow to explain the up-quarks — and the up-quarks are famously top-heavy. That might be fine — we already know the

top quark's mass comes from a different mechanism (it slams into the boundary's ceiling), not from the gentle ladder — but I won't assume it. Elie fires the up-quark mass check blind, and only if the up ladder actually works do we compute the mixing. I also had to correct myself: yesterday I said up and down were the same modes nudged apart; they're not — they're on different-parity shelves entirely, which is a cleaner and more testable picture. One quiet bonus: the even ladder starts at shelf zero, a “flat” mode with no mass — and that's exactly the massless lightest neutrino we already predicted. Suggestive, not banked. Nothing pushed.

Notes only; no toy/theorem claimed (normalization pin + grid split, K1180; corrects F819). F820:  $m = N_c \cdot |Q|$  (T2470 charge=SO(2) weight, T2521 quantized  $1/N_c$ ) → integer weight: down 1, up 2, lepton 3,  $\nu$  0. + F817 parity ( $k \equiv m \pmod{2}$ ) → FORCED grid split: ODD {1,3,5} = down+charged-leptons (m odd); EVEN {0,2,4} = up+neutrinos (m even). ODD ( $\nu$ )<sub>k</sub> = 1:20:840 = down ladder [?](#). EVEN ( $\nu$ )<sub>k</sub> = 1:12:360 (shallow) vs top-heavy up 1:580:78600 → up-mass check OPEN (likely top-ceiling T2529/OP-4/K765, not E1 cascade — Elie's blind fire decides). CORRECTS F819: up is a DIFFERENT parity grid (even), not an in-grid m-shift → mixing is CROSS-grid (up even × down odd). Sequence (K1180): pin normalization [done] → Elie up-mass check BLIND at {0,2,4} → only if clears, run cross-grid mixing fire. Neutrino  $k=0$  = zero-mode hint ( $m_1=0$ /F619 consistency), not banked. @Elie up-mass check precedes mixing; @Cal rule shortfall vs top-ceiling; @Grace weld two-grid split, cross-grid mixing. — Lyra